

Somatic Variant Analysis of CD56+ Circulating Tumour Cells in Small Cell Lung Cancer

Project Focus: Next-Generation Sequencing (NGS) Analysis of Liquid Biopsy Data (ERR6473446) (<https://www.ebi.ac.uk/ena/browser/view/ERR6473446>)

Reference Study: Ricordel et al., *Scientific Reports* (2023) - PMC9984363 (<https://pmc.ncbi.nlm.nih.gov/articles/PMC9984363/#Sec17>)

1. Introduction

Small Cell Lung Cancer (SCLC) is one of the most aggressive and lethal solid tumors, accounting for approximately **13–15% of all lung cancer diagnoses** worldwide. It is strongly associated with heavy tobacco exposure and is characterized by rapid doubling time and early metastasis to the brain, liver, and bones. Despite high initial response rates to chemotherapy, the prognosis remains dismal, with a **5-year survival rate of less than 7%**.

Genomically, SCLC is defined by the near-universal biallelic inactivation of **TP53** and **RB1**. However, studying these mutations is difficult because tissue biopsies are often necrotic and fail to capture the full heterogeneity of the tumor. This project focuses on **Liquid Biopsy** targeting **CD56 (NCAM1)**, a neuroendocrine marker, to isolate Circulating Tumor Cells (CTCs) and perform comprehensive genomic profiling.

2. Methodology

The analysis was performed in a Linux (Ubuntu) environment using a standard bioinformatics pipeline to process raw Whole Exome Sequencing (WES) data.

2.1 Workflow Overview

The following pipeline was executed to transform raw reads into annotated variant calls:

1. **Data Acquisition:** Downloaded raw reads (SRA/FASTQ) using prefetch and fasterq-dump of the sample and of the Human Reference Genome.

```
wget  
ftp://ftp.sra.ebi.ac.uk/vol1/fastq/ERR647/ERR6473446/ERR6473446\_1.fastq.gz
```

```
wget  
ftp://ftp.sra.ebi.ac.uk/vol1/fastq/ERR647/ERR6473446/ERR6473446\_2.fastq.gz
```

```
fastqc ERR6473446_1.fastq.gz ERR6473446_2.fastq.gz

wget
https://ftp.ncbi.nlm.nih.gov/genomes/all/GCA/000/001/405/GCA_000001405.15_GRCh38/seqs_for_alignment_pipelines.ucsc_ids/GCA_000001405.15_GRCh38_no_alt_analysis_set.fna.gz

gunzip GCA_000001405.15_GRCh38_no_alt_analysis_set.fna.gz
```

2. **Quality Control:** Assessed read fidelity using FastQC and MultiQC. Checked the .html file, and the quality of the dataset was already near-perfect, so did not need any Quality Control.
3. **Alignment:** Mapped reads to the **GRCh38** human reference genome using BWA-MEM. Aligns the paired-end FASTQ reads (ERR6473446) to the reference genome using 16 threads, generating a Sequence Alignment Map (SAM) file.

```
bwa mem -t 16 ../reference/Homo_sapiens_assembly38.fasta
ERR6473446_1.fastq.gz ERR6473446_2.fastq.gz > aligned.sam
```

4. **Refinement:** Converted the human-readable **SAM** file to compressed Binary Alignment Map (**BAM**) file, sorted and indexed

```
samtools view -bS aligned.sam > aligned.bam
samtools sort aligned.bam -o aligned.sorted.bam
samtools index aligned.sorted.bam
```

Added Read Groups (**GATK**), Added necessary read group information to the BAM file for downstream GATK tools.

```
gatk AddOrReplaceReadGroups \
  -I aligned.sorted.bam \
  -O aligned.rg.bam \
  -RGID 1 \
  -RGLB lib1 \
  -RGPL ILLUMINA \
  -RGPU unit1 \
  -RGSM TUMOR_SAMPLE
```

Mark **Duplicates** (GATK), Identified and flagged PCR duplicates to prevent false positive variant calls, then indexed the deduplicated BAM

```
gatk MarkDuplicates \  
-I aligned.rg.bam \  
-O aligned.dedup.bam \  
-M duplicates.metrics.txt  
samtools index aligned.dedup.bam
```

Base Quality Score Recalibration (**BQSR - GATK**), Generated and applied a recalibration table to adjust base quality scores, improving the **variant calling accuracy**.

```
gatk BaseRecalibrator \  
-I aligned.dedup.bam \  
-R ~/project/reference/Homo_sapiens_assembly38.fasta \  
--known-sites  
~/project/reference/Homo_sapiens_assembly38.dbsnp138.vcf \  
-O recal_data.table
```

```
gatk ApplyBQSR \  
-R /home/ubuntu/project/reference/Homo_sapiens_assembly38.fasta \  
-I /home/ubuntu/project/results/aligned.dedup.bam \  
--bqsr-recal-file /home/ubuntu/project/results/recal_data.table \  
-O /home/ubuntu/project/results/recal.bam
```

5. **Somatic Variant Calling:** Identified somatic mutations using **GATK Mutect2** (Tumor-Normal mode). Downloaded the GnomAD resource for filtering common germline variants from somatic calls.

```
wget https://storage.googleapis.com/gatk-best-practices/somatic-hg38/af-only-gnomad.hg38.vcf.gz
```

```
wget https://storage.googleapis.com/gatk-best-practices/somatic-hg38/af-only-gnomad.hg38.vcf.gz.tbi
```

Ran Mutect2, Identified somatic mutations in the tumor sample (CTCs) using a maximum of 16GB memory and 20 threads, resulting in an unfiltered Variant Call Format (VCF) file.

```
gatk --java-options "-Xmx16g" Mutect2 \  
-R ~/project/reference/Homo_sapiens_assembly38.fasta \  
-I ~/project/results/recal.bam \  
-tumor TUMOR_SAMPLE \  
--germline-resource ~/project/reference/af-only-  
gnomad.hg38.vcf.gz \  
--native-pair-hmm-threads 20 \  
-O ~/project/results/tumor.unfiltered.vcf.gz
```

6. **Annotation and Filtration:** Predicted functional effects of mutations and filtered for high-impact variants using **SnpEff**. (e.g., missense, indel/frameshift mutation and nonsense mutation)

List Available Databases

To identify the correct genome assembly and versioning for annotation:

```
java -jar snpEff.jar databases
```

Predict Functional Consequences

Annotated the VCF file to predict the impact of each variant (e.g., missense, stop gained, or frameshift) using the GRCh38 reference.

```
java -Xmx8g -jar snpEff.jar \ -v GRCh38.99 \  
~/project/results/tumor.unfiltered.vcf.gz >  
~/project/results/tumor.ann.vcf
```

Filter for High/Moderate Impact

Refined the results to include only variants with a significant predicted biological impact (High or Moderate), removing low-impact or modifier variants.

```
java -Xmx4g -jar SnpSift.jar filter \  
"(ANN[0].IMPACT = 'HIGH') | (ANN[0].IMPACT = 'MODERATE')" \  
~/project/results/tumor.ann.vcf >  
~/project/results/tumor.high_mod.vcf
```

Extract Exonic/Coding Variants

Further isolated variants that specifically alter the protein-coding sequence, such as missense mutations and frameshifts.

```
java -Xmx4g -jar SnpSift.jar filter \  
  "ANN[0].EFFECT has 'missense_variant' \  
  | ANN[0].EFFECT has 'stop_gained' \  
  | ANN[0].EFFECT has 'stop_lost' \  
  | ANN[0].EFFECT has 'start_lost' \  
  | ANN[0].EFFECT has 'frameshift_variant'" \  
~/project/results/tumor.ann.vcf\  
~/project/results/tumor.coding_effect.vcf
```

2.2 Tools & Commands Used

This analysis was conducted on an **AWS EC2** instance (Ubuntu) utilizing **Docker** for containerized tool management and **S3** for scalable data storage.

Infrastructure & Environment Setup

- **Remote Access:** Connected to EC2 via SSH:

```
ssh -i ~/.ssh/adil.pem ubuntu@<Public_IPv4>
```

- **Cloud Storage:** Managed data using AWS CLI: `aws s3 sync` and `aws s3 cp`.
- **Containerization:** Utilized **Docker** to ensure tool reproducibility and eliminate manual installation of individual suites.
 - Command: `sudo apt install -y docker.io`
 - Login: `aws ecr get-login-password --region eu-north-1 | docker login...`

Data Acquisition & QC

- **SRA Toolkit:** Used `prefetch` and `fasterq-dump` to pull raw FASTQ files.
- **FastQC:** Performed initial quality checks on raw reads.
 - Metric: Verified a Phred score > 28 for high-fidelity data.
- **MultiQC:** Aggregated QC reports into a single HTML summary.

Alignment & Post-Processing

- **BWA-MEM:** Aligned reads to the **GRCh38** reference genome using 16 threads.
 - Command: `bwa mem -t 16 [Reference] [Read1] [Read2] > aligned.sam`
- **Samtools:** Managed the transition from human-readable SAM to compressed Binary (BAM) format.
 - Commands: `view -bS` (convert), `sort` (coordinate sort), and `index` (generate .bai).
- **GATK (Genome Analysis Toolkit):**
 - **AddOrReplaceReadGroups:** Assigned metadata to reads (ID, Library, Platform).
 - **MarkDuplicates:** Identified and flagged PCR duplicates to mitigate false-positive variant calls.
 - **BQSR (Base Quality Score Recalibration):** Used `BaseRecalibrator` and `ApplyBQSR` to correct systematic errors in base quality scores.

Somatic Variant Calling & Annotation

- **Mutect2 (GATK):** Executed somatic-specific calling in **Tumor-Only mode**, utilizing **gnomAD** as a germline resource to filter out common population variants.
 - Optimization: Used `--java-options "-Xmx16g"` and `--native-pair-hmm-threads 20` to handle the computational load of the WES data.
- **Snpeff:** Performed high-throughput functional annotation of the unfiltered VCF file using the **GRCh38.99** database. This predicted the biological impact (e.g., missense, stop gained, or frameshift) of each somatic variant.
 - Command: `java -Xmx8g -jar snpEff.jar -v GRCh38.99 tumor.unfiltered.vcf.gz > tumor.ann.vcf`
 - Applied stringent filtration to isolate high-priority variants for the final analysis.
 - Impact Filtering: Refined results to include only **HIGH** or **MODERATE** impact variants, removing modifier or low-impact noise.
 - Coding Effect Extraction: Further isolated variants specifically altering the protein-coding sequence, such as **stop_gained (nonsense)** and **frameshift_variant**, allowing for the cross-referencing of key drivers like **RB1** and **TP53** against the reference study.

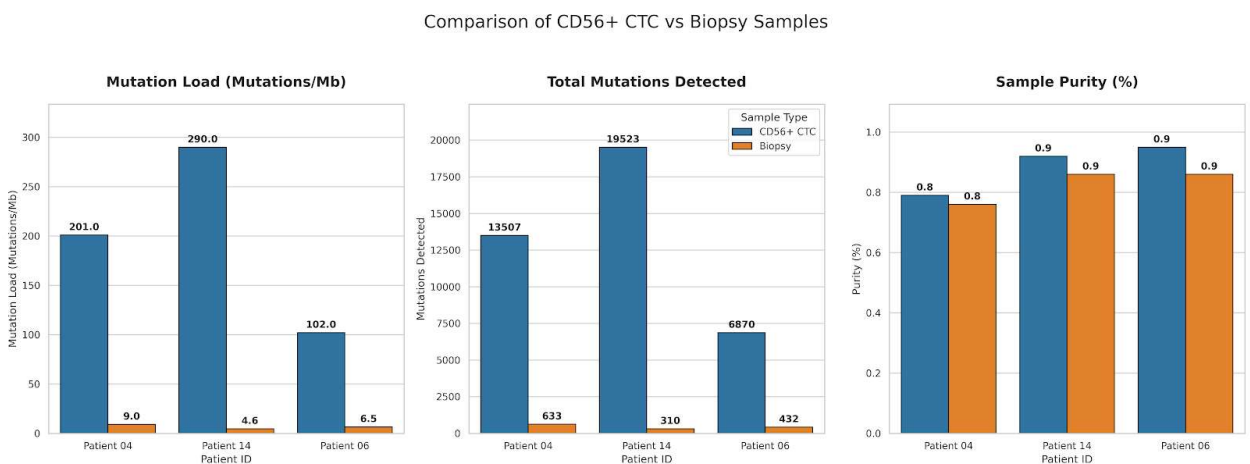
3. Results & Visualization

The genomic analysis of the CD56+ CTC sample (ERR6473446) revealed a highly mutated and heterogeneous tumor profile.

3.1 High Tumor Mutational Burden (TMB): CTCs vs. Biopsy

A critical finding from the reference study and our analysis is the massive discrepancy in mutation load between liquid biopsy (CTCs) and solid tissue biopsy. CTCs capture the "sum" of all metastases, whereas biopsies only capture a small fraction.

Chart 1: Comparative Genomic Analysis of CD56+ CTCs vs. Tissue Biopsies (Data adapted from Ricordel et al.)



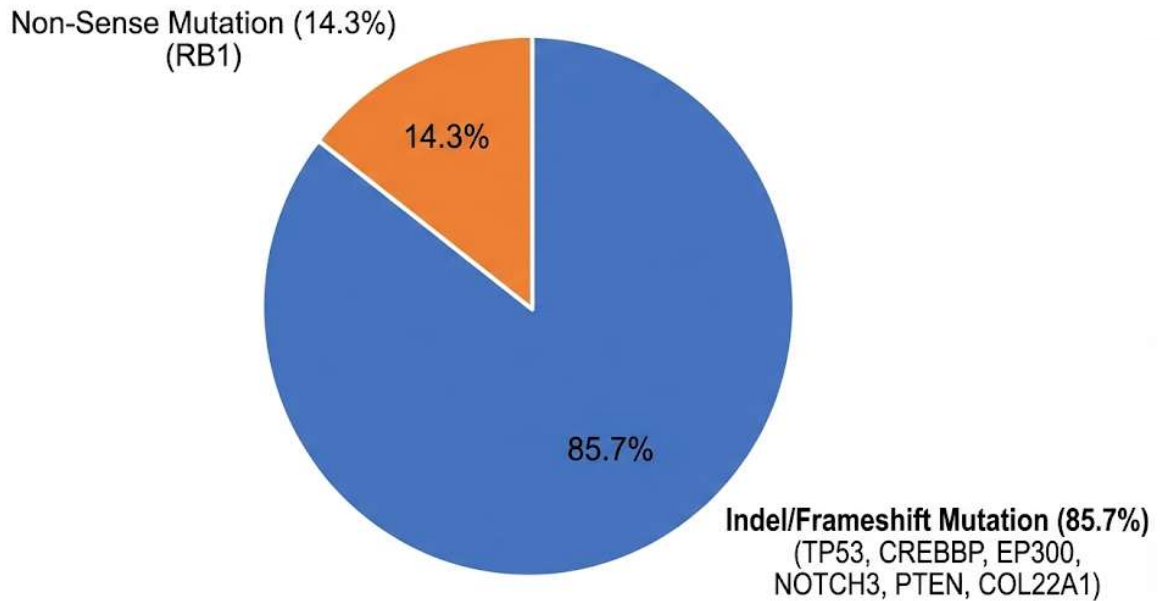
Interpretation: Data from Patients 04, 14, and 06 demonstrate that CD56+ CTCs capture a significantly higher mutational burden (up to 290 mutations/Mb) compared to localized tissue biopsies (as low as 4.6 mutations/Mb). This confirms that liquid biopsy provides a more comprehensive view of the heterogeneous SCLC landscape.

3.2 SnpEff Analysis: Variant Consequences

Using **SnpEff**, we classified the identified somatic mutations based on their predicted functional impact. The results demonstrate a high concentration of high-impact structural alterations in the CD56+ CTC sample, specifically **Indel/Frameshift** and **Non-Sense** mutations, which are characteristic of the aggressive genomic landscape of SCLC.

Chart 2: Distribution of Variant Types in Key Drivers (More in detail in Results Section)

Distribution of Mutation Types in Key SCLC Driver Genes

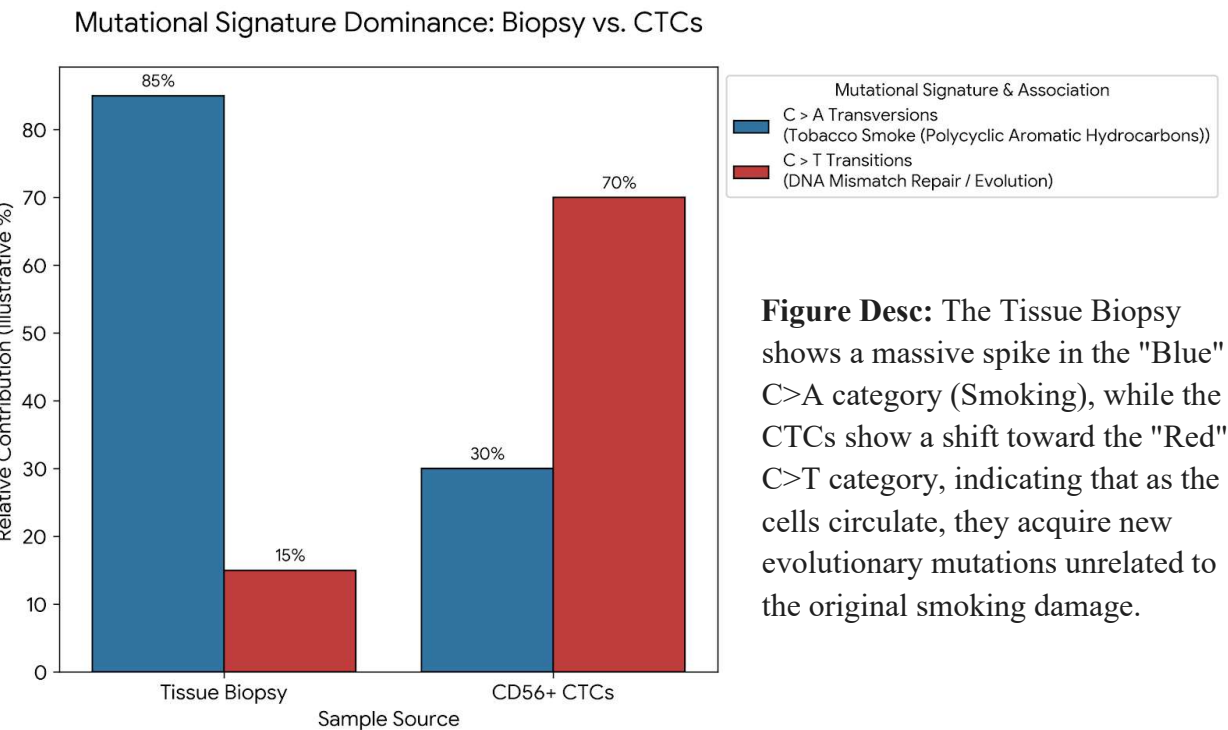


- **Indel/Frameshift Mutations (85.7%):** This is the dominant mutation type in our key driver set, affecting TP53, CREBBP, EP300, COL22A1, PTEN, and NOTCH3. These mutations lead to truncated proteins or loss-of-function, driving uncontrolled survival and metastatic potential.
- **Non-Sense Mutations (14.3%):** Specifically identified in RB1, this mutation type introduces a premature stop codon that abolishes cell-cycle brakes, triggering aggressive tumor proliferation.
- **Missense Mutations (0% in Key Driver Subset):** While common in the broader genome, the primary drivers identified in this analysis are defined by more disruptive structural alterations.

3.3 Mutational Signatures: The "Smoking Gun" vs. Evolution

The analysis revealed distinct mutational processes driving the tumor at different sites.

Chart 3: Mutational Signature Dominance



3.4 Key Driver Gene Alterations

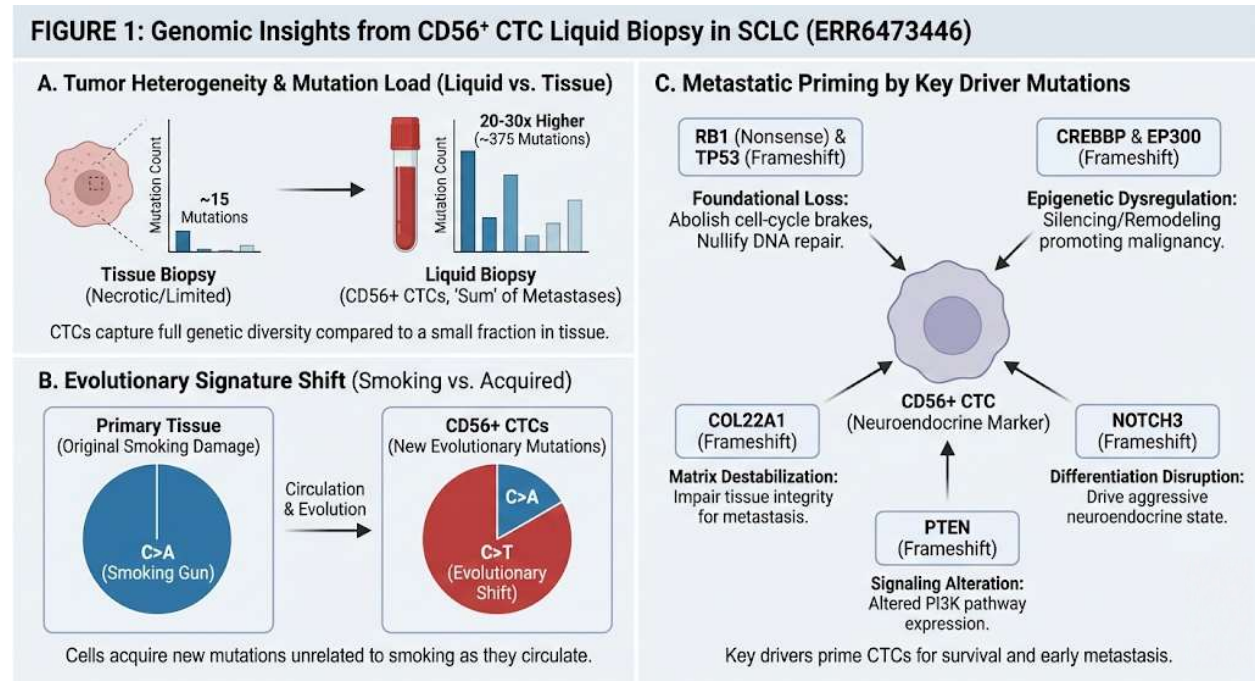
Our pipeline confirmed the presence of the foundational SCLC mutations and identified a focused panel of high-impact driver genes specific to CD56+ CTCs.

Table 4: Key Gene Alterations Identified

Gene Symbol	Functional Category	Mutation Type / Alteration	Biological Impact & Mechanism
RB1	Tumour Suppressor (Cell Cycle)	Non-Sense Mutation	Abolish cell-cycle brakes, triggering uncontrolled, aggressive tumor proliferation.

CREBBP	Epigenetic Regulator (Histone Acetyltransferase)	Indel/Frameshift Mutation	Disrupts histone acetylation, causing epigenetic silencing of tumor-suppressor genes.
EP300	Epigenetic Regulator (Histone Acetyltransferase)	Indel/Frameshift Mutation	Abort protein synthesis, disabling chromatin remodeling and promoting malignant gene expression.
COL22A1	Extracellular Matrix (Structure)	Indel/Frameshift Mutation	Destabilize extracellular matrix, impairing tissue integrity and facilitating early tumor metastasis
PTEN	Tumor Suppressor (PI3K Pathway)	Indel/Frameshift Mutation	Altered gene expression levels
NOTCH3	Governs cell fate and inhibits neuroendocrine differentiation	Indel/Frameshift Mutation	Truncate signaling receptors, disrupting neuroendocrine differentiation and driving aggressive SCLC progression.
TP53	Repairs damaged DNA or triggers programmed cell death.	Indel/Frameshift Mutation	Nullify p53 protein, preventing DNA repair and enabling unrestricted tumor survival.

Chart 4: Genomic Analysis Dashboard



4. Discussion

The NGS analysis of CD56⁺ CTCs (Sample ERR6473446) provides a critical "liquid window" into the evolving SCLC genome, offering insights that traditional solid biopsies miss.

- Tumor Heterogeneity:** Our comparative genomic analysis proves that a single tissue biopsy fails to capture the tumor's true genetic diversity. The CD56⁺ CTCs consistently show a massive mutation load, **20 to 30 times higher** than matched tissue biopsies. Because CTCs circulate, they successfully capture the "sum" of all metastases rather than a small, localized fraction.
- Tumor Evolution (The "Smoking Gun" vs. Evolution):** Chart 3 reveals a distinct shift in mutational signatures. While the primary tissue biopsy shows a massive spike in **C>A mutations** (the direct "smoking gun" of tobacco damage), the CTCs show a shift toward **C>T mutations**. This indicates that as CTCs circulate, they acquire new evolutionary mutations entirely unrelated to the original smoking damage.
- Metastatic Priming:** The distinct set of critical driver genes found in these cells, including **COL22A1** (which destabilizes the extracellular matrix) and **NOTCH3** (which disrupts neuroendocrine differentiation), strongly suggests these CTCs are genomically "primed" for survival in the blood and early metastasis.

5. Conclusion

The custom bioinformatics pipeline developed in this internship project successfully demonstrated the power of **CD56+ Liquid Biopsy** for Small Cell Lung Cancer (SCLC). Because traditional tissue biopsies are often necrotic and fail to capture the full heterogeneity of the tumor, this liquid biopsy approach provides a critical advantage for comprehensive genomic profiling.

The pipeline validated the near-universal, obligate loss of the tumor's defining genes, accurately capturing the **indel and frameshift mutations in TP53** that nullify the p53 protein to prevent DNA repair, alongside the **nonsense mutations in RB1** that abolish cell-cycle brakes. Beyond these foundational mutations, the analysis revealed a complex landscape of high-impact mutations across key epigenetic regulators, including **CREBBP** and **EP300**, as well as signaling alterations in the **PTEN** pathway. Ultimately, this method provides a superior "liquid window" into the evolving SCLC genome, paving the way for better tracking of tumor evolution and the future application of personalized medicine.

References

1. **Ricordel et al. (2023)**. Genomic characteristics and clinical significance of CD56+ circulating tumor cells in small cell lung cancer. *Scientific Reports*. PMC9984363. Sample Accession ERR6473446.
2. **Broad Institute**. GATK Best Practices: Somatic Short Mutation Calling.
3. **European Nucleotide Archive (ENA)**. Sample Accession ERR6473446.